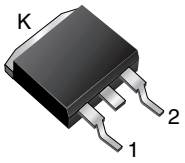
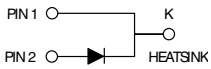
Schottky Barrier Rectifier

D²PAK (TO-263AB)





PRIMARY CHARACTERISTICS	
$I_{F(AV)}$	16 A
V_{RRM}	45 V, 60 V
I_{FSM}	150 A
V_F	0.57 V, 0.65 V
T_J max.	150 °C
Package	D²PAK (TO-263AB)
Circuit configuration	Single

FEATURES

RoHS COMPLIANT
HALOGEN FREE Available

- Power pack
- Guardring for overvoltage protection
- Low power loss, high efficiency
- Low forward voltage drop
- High forward surge capability
- High frequency operation
- Meets MSL level 1, per J-STD-020, LF maximum peak of 245 °C
- AEC-Q101 qualified available
  - Automotive ordering code: Base P/NHM3
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

TYPICAL APPLICATIONS

For use in low voltage, high frequency rectifier of switching mode power supplies, freewheeling diodes, DC/DC converters, and polarity protection application.

MECHANICAL DATA

Case: D²PAK (TO-263AB)

Molding compound meets UL 94 V-0 flammability rating

Base P/N-M3 - RoHS-compliant, halogen-free, commercial grade

Base P/NHM3 - RoHS-compliant, halogen-free, AEC-Q101 qualified

Terminals: matte tin plated leads, solderable per J-STD-002 and JESD 22-B102

M3 suffix meets JESD 201 class 1A whisker test, HM3 suffix meets JESD 201 class 2 whisker test

Polarity: as marked

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MAXIMUM RATINGS (T_C = 25 °C unless otherwise noted)				
PARAMETER	**SYMBOL**	**MBRB1645**	**MBRB1660**	**UNIT**
Maximum repetitive peak reverse voltage	V_{RRM}	45	60	V
Working peak reverse voltage	V_{RWM}	45	60	
Maximum DC blocking voltage	V_{DC}	45	60	
Maximum average forward rectified current at T_C = 125 °C	$I_{F(AV)}$	16		A
Peak forward surge current 8.3 ms single half sine-wave superimposed on rated load	I_{FSM}	150		
Peak repetitive reverse current at t_p = 2.0 µs, 1 kHz	I_{RRM}	1.0	0.5	
Voltage rate of change (rated V_R)	dV/dt	10 000		V/µs
Operating junction temperature range	T_J	-65 to +150		°C
Storage temperature range	T_{STG}	-65 to +175		

ELECTRICAL CHARACTERISTICS (T_C = 25 °C unless otherwise noted)						
PARAMETER	**SYMBOL**	**TEST CONDITIONS**		**MBRB1645**	**MBRB1660**	**UNIT**
Maximum instantaneous forward voltage	V_F (1)	I_F = 16 A	T_C = 25 °C	0.63	0.75	V
		I_F = 16 A	T_C = 125 °C	0.57	0.65	
Maximum instantaneous reverse current at DC blocking voltage	I_R (1)	Rated V_R	T_C = 25 °C	0.2	1.0	mA
			T_C = 125 °C	40	50	

Notes
(1) Pulse test: 300 µs pulse width, 1 % duty cycle
(2) Pulse test: pulse width ≤ 40 ms

THERMAL CHARACTERISTICS (T_C = 25 °C unless otherwise noted)			
PARAMETER	**SYMBOL**	**MBRB**	**UNIT**
Typical thermal resistance from junction to case	$R_{\theta JC}$	1.5	°C/W

ORDERING INFORMATION (Example)					
PACKAGE	**PREFERRED P/N**	**UNIT WEIGHT (g)**	**PACKAGE CODE**	**BASE QUANTITY**	**DELIVERY MODE**
D^2PAK (TO-263AB)	MBRB1645-M3/I	1.33	I	800/reel	Tape and reel
D^2PAK (TO-263AB)	MBRB1645HM3/I (1)	1.33	I	800/reel	Tape and reel

Note
(1) AEC-Q101 qualified

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RATINGS AND CHARACTERISTICS CURVES (T_C = 25 °C unless otherwise noted)

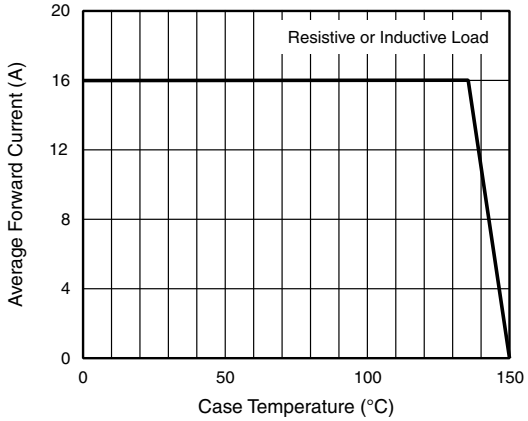


Fig. 1 - Forward Current Derating Curve

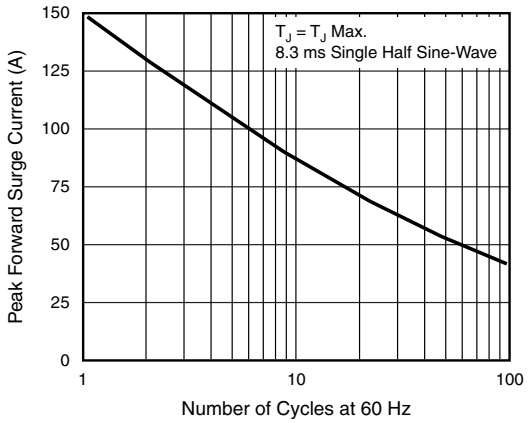


Fig. 2 - Maximum Non-Repetitive Peak Forward Surge Current

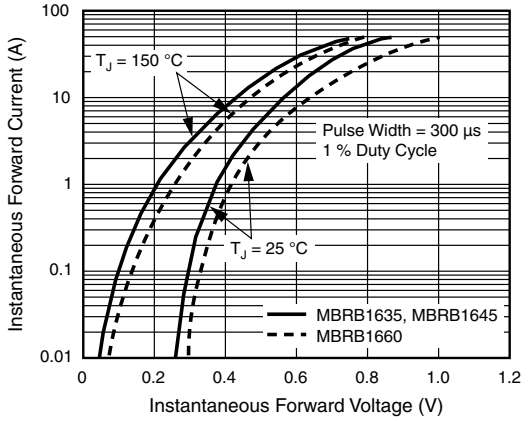


Fig. 3 - Typical Instantaneous Forward Characteristics

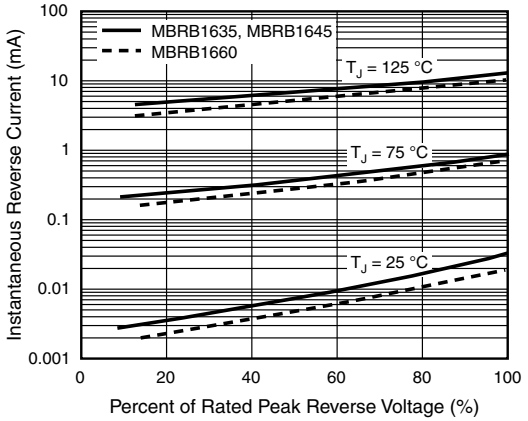


Fig. 4 - Typical Reverse Characteristics

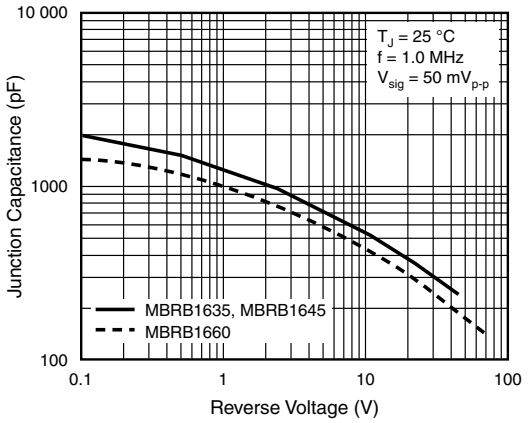


Fig. 5 - Typical Junction Capacitance

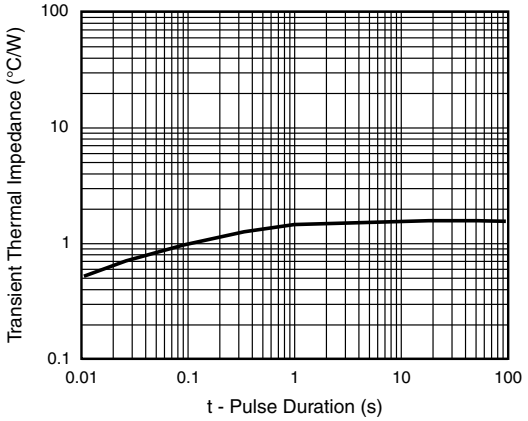
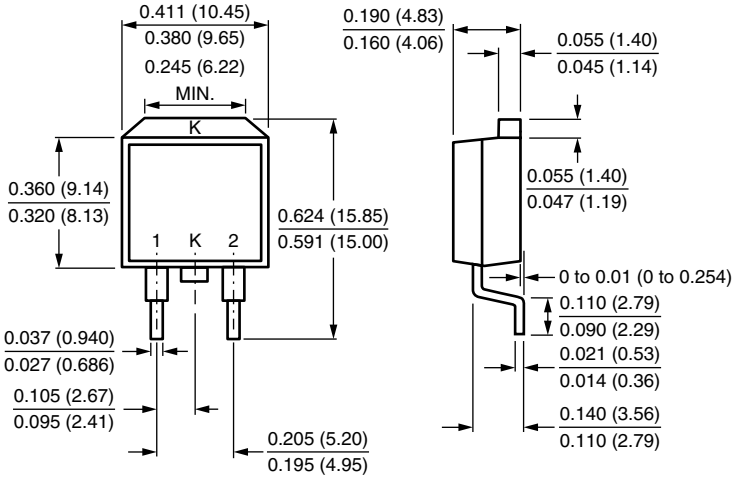


Fig. 6 - Typical Transient Thermal Impedance

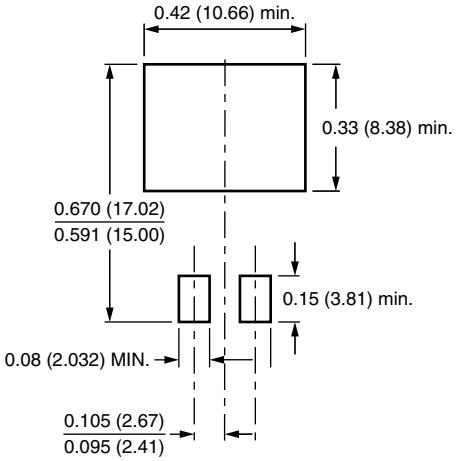
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PACKAGE OUTLINE DIMENSIONS in inches (millimeters)

D^2PAK (TO-263AB)



Mounting Pad Layout



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